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Enhancing clinical law education through immersive virtual reality: A flow experience perspective

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ABSTRACT

Background: The existing literature lacks experimental studies that provide a genuine virtual reality experience, particularly within the context of the flow experience, as it relates to clinical legal education.
Aims: This study investigates the impact of immersive Virtual Reality (VR) on clinical law education, with a particular focus on the flow experience.
Sample: Eighty-three law students participated in case practices.
Method: The study adopted an experimental design that included a random post-test control group, two distinct VR environments, immersive VR and desktop VR, were employed to create authentic scenarios mirroring real-life legal cases, and differences between the groups were tested using One-Way Multivariate Analysis of Variance (MANOVA).
Results: The results revealed that immersive VR significantly enhances the flow experience of law students compared to desktop VR. Participants in the immersive VR group demonstrated higher levels of focused attention, telepresence, time distortion, and interaction, indicating a more profound immersion. Surprisingly, there was no significant difference in terms of enjoyment between the two groups, contradicting previous research in this area.
Conclusions: This study is pioneering in its approach, offering a unique contribution to the field of law education. It highlights the potential of immersive VR as a valuable tool for enhancing clinical legal education and creating more engaging learning experiences. Expanding the use of these virtual environments to both practical and theory-based learning methods presents an exciting avenue for further exploration in the field of legal education.

1. Introduction

Clinical law education, which is integrated into the law programs of many higher education institutions around the world, constitutes one of the most fundamental pedagogical pillars of legal education (Evans et al., 2017; Giddings, 2013). Clinical legal education, commonly employed in traditional face-to-face law schools, refers to experiential learning methods that engage students in public practical activities (Jones et al., 2017; Maharg, 2017). Clinical legal education seeks to prepare law students for the practice of law by providing them with the practical skills, ethical principles, and professional values necessary to be effective and responsible attorneys, prosecutors and judges. One of the primary goals of clinical legal education is to provide law students with practical skills essential for legal practice (Giddings, 2013). Clinical programs offer students the opportunity to apply legal theory in real-world contexts. This hands-on experience helps bridge the gap between classroom learning and actual legal practice. With clinical legal education, which has the capacity to both broaden and deepen learning, students benefit from sustainable experience through reflective practices (Evans et al., 2017).

Given that clinical legal education is inherently practical, it presents opportunities for both service and teaching through the utilization of various technological tools (Barry et al., 2000). Among these tools, virtual reality technologies stand out as having the highest potential, as

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they offer the capacity to deliver genuine real-life experiences. Although virtual reality provides immersive and realistic simulations of legal scenarios, allowing law students to gain practical experience and skills in a controlled and safe environment, the number of studies conducted in this context is quite limited. This lack of studies on utilizing immersive virtual reality, which provides valuable data for evaluation by tracking students' actions and decisions, and supporting students to enter legal practice with higher readiness and self-confidence by providing feedback to help students improve their legal reasoning and judgement, is a critical gap in the field. Furthermore, the existing literature lacks experimental studies that provide a genuine virtual reality experience, particularly within the context of the flow experience, as it relates to clinical legal education. While previous research has delved into areas such as online virtual legal clinics (McCrimmon et al., 2016; Ryan, 2020), legal education utilizing virtual reality technology on smartphones (Ryan, 2020), and the implementation of virtual courts in legal education (Öngöz et al., 2017; Rogers, 2016), there is a notable absence of research exploring the immersive virtual reality environment and its impact on the flow experience within the realm of legal education.

In this study, the concept of flow experience is conceptualized as a multi-dimensional construct, encompassing a range of elements and experiences, rather than being viewed as a simplistic one-dimensional concept. Building upon extensive research within the realm of flow experience from prior studies (e.g., Hoffman & Novak, 2009; Kwak et al., 2014; Wu & Wang, 2011), it was hypothesized that a successful flow experience within immersive virtual reality environments should comprise five fundamental components: enjoyment, focused attention (concentration), telepresence, time distortion, and interaction. In this context, the primary objective of this study is to investigate the impact of immersive virtual reality environments on the flow experience of law school students, particularly within the domain of information technology law (IT law - Cyberlaw). To achieve this goal, an experimental research design was employed, aiming to establish causal comparisons related to the flow experience. However, it was anticipated that directly comparing the virtual reality environment with traditional clinical practices in the context of flow experience might yield less meaningful and robust results. As a result, it was deemed appropriate to develop two distinct versions of the virtual reality environment, namely immersive virtual reality and desktop virtual reality, for use in clinical case applications.

2. Theoretical background

2.1. Immersive virtual reality and flow experience

Virtual reality refers to technological environments that enable users to immerse themselves in simulated real-life situations, allowing for both physical and mental interaction within these environments. This immersion is facilitated through the use of wearable technologies equipped with various sensors (Doğan, 2021; Doğan & Şahin, 2024). One central concept associated with virtual reality is "immersion," which characterizes a user's deep engagement in a virtual environment, often to the extent that their awareness of time and the physical world is temporarily suspended, resulting in a strong sense of "presence" within the virtual context (Radianti et al., 2020). This term, as defined by Freina and Ott (2015), signifies the perception of physical existence within a non-physical realm, achieved by surrounding the user with images, sounds, or other stimuli in a virtual reality system. Virtual reality experiences can be accessed through various devices, including desktop computers, immersive headsets, or sophisticated CAVE (Cave Automatic Virtual Reality Environment) systems (Butussi & Chittaro, 2018). The key distinguishing factor between a virtual reality learning session accessed through a viewing headset or CAVE system as opposed to one accessed via a desktop computer is the level of immersion. Put differently, virtual reality applications that utilize viewing headsets offer a significantly higher level of immersion compared to those deployed on desktop configurations. Furthermore, the literature underscores the inherent relationship between immersion and the flow experience. Studies have consistently shown that users engaged in immersive virtual reality environments tend to experience a much higher degree of flow compared to those using desktop applications (Pallavicini et al., 2019).

Flow Theory, initially proposed by Mihaly Csikszentmihalyi in the 1970s, has found extensive application in contemporary information technologies. This theory is rooted in four fundamental components: internal interest, curiosity, focused attention, and control. According to Csikszentmihalyi (1990), Flow Theory can be applied to various facets of life improvement, defining "flow" as a pleasurable experience characterized by a heightened sense of behavioral control, happiness, and enjoyment. Flow entails a state of deep engagement where nothing else appears to be of significance (Csikszentmihalyi, 1990). In essence, it represents an optimal experience where individuals derive self-satisfaction from their activities (Kwak et al., 2014).

An examination of the existing literature reveals that previous research has identified time distortion and telepresence as key variables influencing the flow experience (Bridges & Florsheim, 2008; Hoffman & Novak, 2009; Kwak et al., 2014). Furthermore, findings from flow studies have underscored the critical roles of enjoyment and interaction in shaping flow experiences (Chen et al., 2008; Kwak et al., 2014; Wu & Wang, 2011). Similarly, the foremost parameters associated with the flow experience within computer-mediated environments encompass variables like change, skill, goals, and experiential dimensions like enjoyment, concentration, time distortion, loss of personal consciousness, and telepresence (Finneran & Zhang, 2005). One of the primary features of virtual reality applications is their ability to evoke a sense of presence in the user, which is a crucial component of the flow experience (Mikropoulos & Strouboulis, 2004; Novak et al., 2000; Takatalo et al., 2008). For users to experience flow in virtual environments, Reid (2004) suggested that the entertainment factor is crucial. Cheng et al. (2014) emphasized the importance of the interaction factor, while Mütterlein (2018) highlighted the significance of the feeling of being there. Research indicates that time distortion, interaction, focused attention, telepresence, and enjoyment are crucial for achieving the flow experience. These elements are associated with the immersion state of virtual reality (Freina & Ott, 2015; Radianti et al., 2020). In essence, the degree of immersion in a virtual environment is believed to influence variables such as detachment from the real environment, concentration on the virtual environment, enjoyment within it, and interaction with it— all of which are essential components of the flow experience. Consequently, in this study, enjoyment, focused attention (concentration), telepresence, time distortion, and interaction constructs have been identified as precursors to the flow experience.

The study formulated the following hypotheses to investigate the flow experience in immersive and desktop VR environments using these five constructs.

H1. The use of immersive VR in IT law clinics fosters a more pronounced flow experience compared to desktop VR.

H1a. Immersive VR employed in IT law clinics leads to a higher level of focused attention compared to desktop VR.

H1b. Immersive VR employed in IT law clinics leads to a higher level of time distortion compared to desktop VR.

H1c. Immersive VR employed in IT law clinics leads to a higher level of telepresence compared to desktop VR.

H1d. Immersive VR employed in IT law clinics leads to a higher level of interaction compared to desktop VR.

H1e. Immersive VR employed in IT law clinics leads to a higher level of enjoyment compared to desktop VR.

3. Scenario design & virtual reality development

During the initial design and development phase for the research's implementation, we undertook the task of designing case scenarios and crafting the virtual reality (VR) environment. In the scenario design, we collaborated with both an IT law expert and a communication field expert to outline the scenario, characters, sounds, animations, and the overall environment within which participants would immerse themselves in the VR environment. These scenarios were constructed as flexible, branching structures, adaptable to the user's preferences and decisions as they progressed through the cases. Real-life scenarios were designed to follow different paths based on the law student's choices, allowing for diverse experiences under various conditions, with the ultimate goal of enhancing the effectiveness of clinical practice. Additionally, voice-overs for the characters' dialogues were recorded during this stage.

The second phase encompasses the design and development of the virtual reality (VR) environment. Utilizing the user interface tools within the Unity editor, various interface elements such as text boxes and buttons were meticulously designed to provide users with essential information and interactions. Throughout this phase, a seamless user experience was ensured by consistently assessing the application's functionality via the Unity editor's playback mode. Upon successfully establishing the interaction between the interface elements and the predefined scenarios, specific libraries tailored for Oculus Rift were integrated into the application. This crucial step facilitated the transition of the application from a conventional monitor screen to the immersive Oculus Rift environment. It's worth noting that the entire application development process was conducted using the C# programming language.

The virtual environment is meticulously designed to replicate a courtroom setting (Fig. 1). Within this court simulation, which comprises four distinct hearings, various roles are assigned, including the plaintiff, the defendant, lawyers representing both parties, the clerk, and the judge. Notably, the participant assumes the pivotal role of the judge, responsible for overseeing the case proceedings. Upon commencing the simulation, participants encounter an informative text displayed on the screen. After perusing this text, participants are prompted to interact with the interface, setting the simulation in motion. Subsequently, participants are presented with choices that significantly influence the course of the case. By carefully reading the status updates on the screen, participants can exercise their judgment and tailor the case's progression as they see fit. Following the decision-making phases, participants primarily assume an observational role, attentively listening to the case as it unfolds. Throughout this process, various characters, including the lawyers, plaintiff, defendant, and judge, engage in dialogues. Given the importance of grasping the case's context solely through experiencing the environment, it is imperative that participating students have prior knowledge from the IT law course. Consequently, this project involved 3rd and 4th-year law school students who possessed the requisite background.

In the immersive VR environment, participants assume the role of a judge, experiencing the courtroom from a first-person perspective in FPS (First-Person Shooter) mode (Fig. 2). Equipped with a controller in each hand, they navigate the interface and flip through pages using these controllers. The simulation grants participants the ability to interact with the environment by manipulating these controllers, facilitating a dynamic and immersive experience. Participants enjoy a 360-degree field of view, enabling them to explore the courtroom from various angles. They can move both horizontally and vertically within the judge's bench area, lean down to observe the clerk in front, and have a comprehensive view of both the plaintiff and defendant. Conversely, in the desktop VR version, the courtroom layout mirrors that of the VR environment. Participants, once again, embody the role of the judge in a first-person perspective. However, control in this version is managed through a mouse, allowing participants to gaze around the environment. It's worth noting that, unlike the VR experience, participants in the desktop application are stationary at the judge's bench and cannot physically relocate within the courtroom. After progressing through the reading texts with the mouse's assistance, participants engage with voiceovers and subsequently make their decisions at the conclusion of each case.

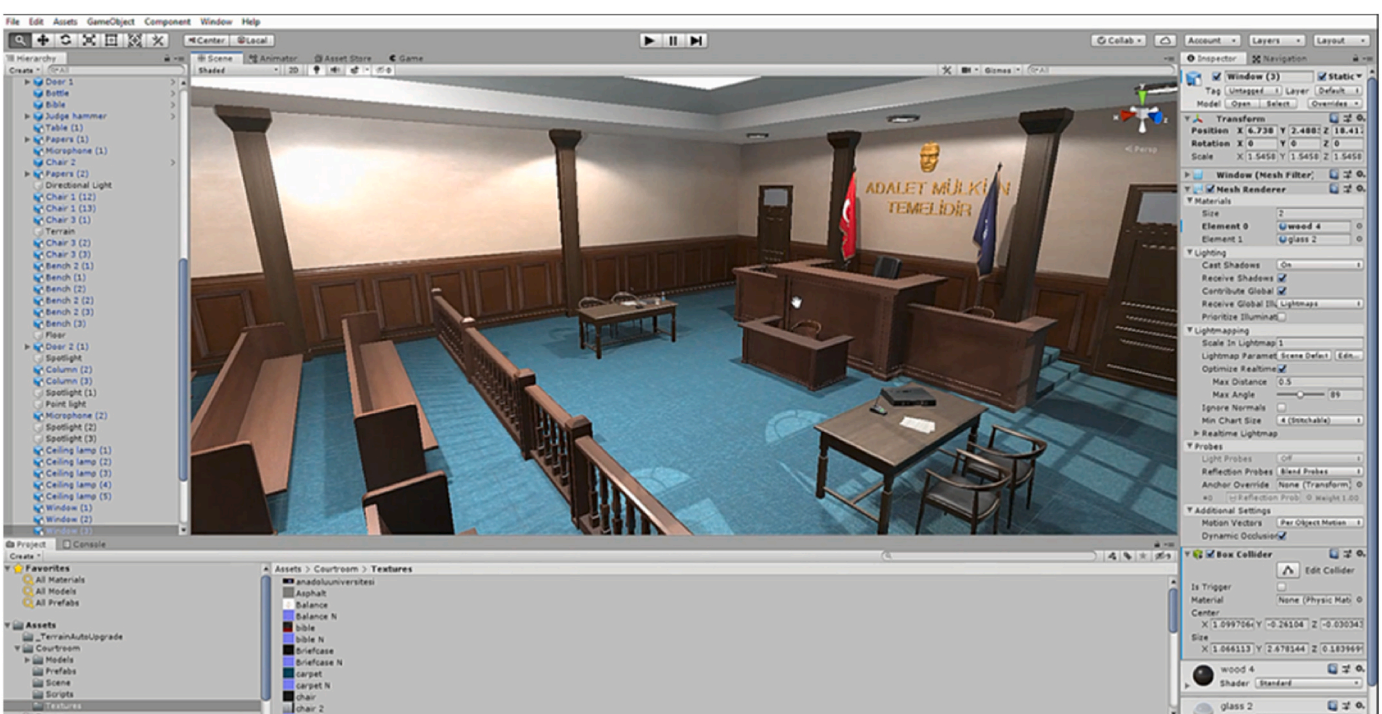

Fig. 1. Virtual reality environment - Unity.

Fig. 2. Immersive virtual reality and desktop virtual reality.

4. Methodology

4.1. Research design

The study employed an experimental design featuring a random post-test control group, a well-regarded approach within experimental research. This design is often regarded as the optimal choice for experimental studies, especially when each group consists of a minimum of 40 participants (Fraenkel et al., 2012). The schematic representation of the experimental design involving the random post-test control group is provided in Table 1.

The participants were assigned randomly to either the experimental group or the control group (Table 1). In the experimental group, clinical practice was conducted using immersive virtual reality, while the control group utilized desktop virtual reality. Following the completion of the application process, measurements were taken to assess the participants' levels of interaction, telepresence, focused attention, time distortion, and enjoyment. The study then analyzed the differences in post-test scores obtained from these measurements between the experimental and control groups.

An approval from a university ethics committee was obtained prior to the implementation of the study.

Table 1
Experimental design with random post-test control group.

Group	N	Application	Measures
Experimental group	R (41)	Immersive Virtual Reality	Interaction Telepresence Time distortion Focused attention Enjoyment
Control group	R (42)	Dekstop Virtual Reality	Interaction Telepresence Time distortion Focused attention Enjoyment

4.2. Participants

During the tool adaptation phase of the study, a convenient sampling method was employed, leading to the inclusion of 435 law students with prior virtual reality experience from a state university. Among these participants, 180 were women (41.4%), and 255 were men (58.6%). The participants' average age was 34.20. According to Kline (2010), having a sample size ranging from 100 to 200 participants is considered sufficient for conducting Confirmatory Factor Analysis (CFA).

In the subsequent experimental research phase, a convenient sampling method was once again used to select participants for both the experimental and control groups. In total, 83 law students were chosen to partake in the research and were randomly assigned to their respective groups as per the experimental design. The demographic information of the participants can be found in Table 2.

4.3. Data collection tool

The study utilized a measurement tool consisting of 5 factors and 16 items designed to assess various dimensions of the flow experience. The factors within the tool encompassed focused attention (3 items), telepresence (4 items), time distortion (3 items), enjoyment (3 items), and interaction (3 items). To develop this measurement tool, items related to focused attention, telepresence, and time distortion were adapted from Kwak et al. (2014), items related to enjoyment were adapted from Chen et al. (2008) and Kwak et al. (2014), while items concerning interaction were adapted from Hudson et al. (2019).

The process of adapting and establishing the construct validity of the measurement tool followed several steps. Initially, the translation of the

Table 2
Demographic profile of the students.

VR Type			*f*	%
Immersive Virtual Reality	Gender	Female	24	58.5
		Male	17	41.5
	Competence	Low	4	9.8
		Moderate	19	46.3
		High	18	43.9
Av. Age: 21,98				
Dekstop Virtual Reality	Gender	Female	19	45.2
		Male	23	54.8
	Competence	Low	5	11.9
		Moderate	23	54.8
		High	14	33.3
Av. Age: 22,1				

scale items was performed by language experts, ensuring their compatibility with the original items. Subsequently, an evaluation of content validity was conducted with input from information technologies field experts. After receiving feedback from these experts, a pilot application was carried out involving 20 law students. This pilot aimed to assess the clarity of the items and determine the average completion time of the tool.

The first phase of data collection aimed to assess the construct validity of the scale through confirmatory factor analysis (CFA). CFA is a statistical technique employed to assess the degree to which a scale aligns with the dataset, to determine the consistency with which factors measure their intended variables, and to validate the factor structure of the scale (Şahin, 2021; Ursavaş, 2014). In the CFA process conducted within the study, various parameters were examined, including item loadings, Cronbach's alpha (α), composite reliability (CR), average variance extracted (AVE) (Table 3, Fig. 3), and model fit indices (χ2/sd, RMSEA, SRMR, NFI, TLI, CFI).

In accordance with the literature, ideal values for construct validity include item loadings above 0.6, Cronbach's alpha (α) and composite reliability above 0.70, and AVE above 0.5 (Hair et al., 2011, 2017; Hu & Bentler, 1999). The results indicated that the item loadings for the factors ranged from 0.674 to 0.902, all exceeding the 0.6 threshold. This ensured the reliability of the scale at the item level. Regarding Cronbach's alpha (α) and composite reliability, values ranged between 0.848 and 0.890, and 0.849 and 0.891, respectively, all surpassing the 0.7 benchmark. This suggests that the scale's reliability meets these criteria effectively. Lastly, the average variance values extracted for the factors ranged from 0.622 to 0.732, all exceeding the 0.5 threshold, indicating that the scale's dimensions effectively and consistently measure the intended variables (Table 3) (Hair et al., 2011, 2017).

The model's fit indices were assessed using Confirmatory Factor Analysis (CFA). Fit indices such as NFI (Normed Fit Index), TLI (Tucker-Lewis Index), CFI (Comparative Fit Index), χ2/df (Chi-square to degrees of freedom ratio), SRMR (Standardized Root Mean Square Residual), and RMSEA (Root Mean Square Error of Approximation) were examined, and their values were compared to recommended thresholds established in the literature (see Table 4).

Table 3
Construct validity results.

Factors	Items	Item Loadings	α	CR	AVE
Focused Attention	FA1	0,797	0,848	0,849	0,652
	FA2	0,764			
	FA3	0,858			
Telepresence	TP1	0,809	0,856	0,867	0,622
	TP2	0,773			
	TP3	0,881			
	TP4	0,680			
Time Distortion	TD1	0,882	0,848	0,858	0,671
	TD2	0,883			
	TD3	0,674			
Enjoyment	ENJ1	0,843	0,890	0,891	0,732
	ENJ2	0,885			
	ENJ3	0,839			
Interaction	INTR1	0,742	0,880	0,885	0,722
	INTR2	0,896			
	INTR3	0,902			

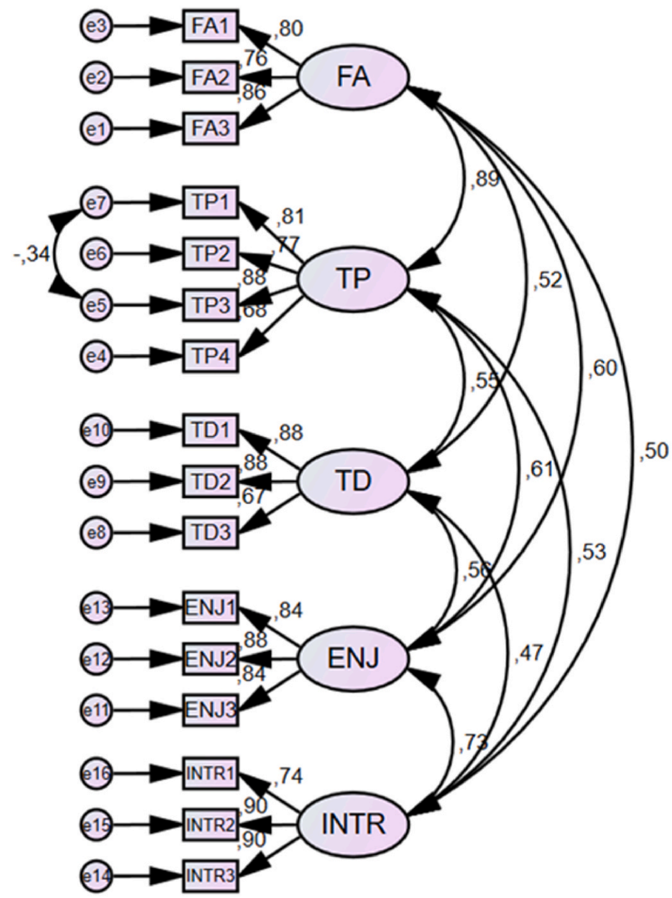

Fig. 3. Confirmatory factor analysis.

The model fit results indicate that the scale's fit values fall within excellent and appropriate ranges (χ2/df = 3.150, RMSEA = 0.070, SRMR = 0.049, NFI = 0.940, TLI = 0.945, CFI = 0.957). These findings suggest that the scale exhibits a very good model fit (Table 4). Based on the outcomes of the CFA conducted in this study, it can be concluded that the scale demonstrates strong construct validity, and the factor structure has been substantiated. Consequently, it can be inferred that the scale presents a valid and reliable instrument for investigating variables related to the virtual reality technology experiences of law students.

Table 4
Model fit.

Fit Index	Threshold	Fit Values	Source	Results
χ2/sd	0≤ χ2/sd ≤ 5	3150	Sümer (2000)	Excellent
SRMR	0≤SRMR≤0,05	0,049	Kline (2011)	Excellent
RMSEA	0≤RMSEA≤0,06	0,070	Thompson (2004)	Acceptable
NFI	0,95≤CFI≤1	0,940	Thompson (2004)	Good
TLI	0,90≤TLI≤1	0,945	Schumacker and Lomax (1996)	Excellent
CFI	0,95≤CFI≤1	0,957	Hu and Bentler (1999)	Excellent

χ2 = 292,945; sd = 93.

4.4. Data analysis

For the analysis of the quantitative data gathered during the experimental phase of the research, a One-Way Multivariate Analysis of Variance (MANOVA) was employed. MANOVA provides enhanced statistical power by considering interactions between dependent variables, thereby reducing potential errors (Can, 2014). Before conducting MANOVA, the necessary prerequisites were carefully assessed.

In the study, MANOVA was employed to compare the outcomes related to the variables of interaction, telepresence, enjoyment, time distortion, and focused attention. Prior to conducting MANOVA, several assumptions were rigorously examined. These assumptions encompassed the evaluation of homogeneity of variance-covariance matrices, assessment of normal distribution, scrutiny for multicollinearity, and identification of singularity issues. To enhance the data quality and ensure multivariate normal distribution, Mahalanobis distance, a standard score, was employed to detect and address extreme values. Moreover, the skewness and kurtosis values, computed to assess the distribution of the variables, fell within the range of −2 to +2 (Table 5). These values align with the acceptable thresholds for the assumption of normal distribution, as outlined by George and Mallery (2010).

The assessment of multicollinearity and singularity involved an examination of the correlations between the variables. The results revealed that the correlation coefficients among the dependent variables were predominantly below 0.90 and averaged around 0.60 (Table 6). This indicates a multicollinear relationship between the variables with no singularity problem present (Pallant, 2007). Consequently, the assumptions concerning multicollinearity and singularity were upheld. However, it was found that the assumption of homogeneity of variance-covariance matrices was not met. To verify this condition, the results of Box's covariance matrix equality test were scrutinized ($p < 0.05$). Tabachnick and Fidell (2012) recommend the use of the Pillai's Trace method in MANOVA analysis when the assumption of variance-covariance homogeneity is violated. Therefore, Pillai's Trace value was considered when interpreting the results derived from MANOVA. In light of these findings, it was evident that there were no issues related to the necessary prerequisites for conducting MANOVA. Therefore, the analysis proceeded as planned.

5. Results

The examination of whether there existed a distinction in the scores of participants between the immersive VR and desktop VR groups on the relevant variables was conducted via MANOVA (Table 7). The results reveal a significant difference in the participants' scores related to interaction, telepresence, enjoyment, time distortion, and focused attention between the immersive VR and desktop VR groups (Pillai's Trace = 0.175, $F(5,77) = 3.269$, $\eta 2 = 0.175$, $p < .05$). The virtual reality environment exerts a substantial influence on the relevant variables ($\eta 2 > 0.14$), and the sample size achieved is sufficient for making comparisons based on the power value (power >0.80) (Cohen, 1988). In light of this finding, it can be concluded that the H1 hypothesis is supported.

In the subsequent phase, ANOVA results were scrutinized to identify the specific dependent variables that exhibited significant differences as indicated by the MANOVA outcomes (Table 8). To mitigate the risk of committing a type one error during the interpretation of ANOVA results, the critical significance threshold was set at 0.0125 (0.05/4) using Bonferroni correction.

Upon examination of Tables 8 and it becomes apparent that the immersive virtual environment significantly influences focused attention (FFA $(1,81) = 12.576$, $p < 0.0125$), telepresence (FTEL $(1,81) = 10.082$, $p < 0.0125$), time distortion (FTD $(1,81) = 8.262$, $p < 0.0125$), and interaction (FINT $(1,81) = 9.201$, $p < 0.0125$). However, for enjoyment (FENJ $(1,81) = 5.220$, $p > 0.0125$), there is no significant effect attributed to immersive VR. The impact of the virtual environment on focused attention, telepresence, time distortion, and interaction are characterized as moderate ($0.06<\eta 2<0.14$). Consequently, hypotheses H1a, H1b, H1c, and H1d are supported. Pairwise comparison tests were conducted to identify specific groups within the independent variable that exhibited significant differences (Table 9). As per Tables 9 and it is evident that immersive virtual reality elicits greater focused attention, telepresence, time distortion, and interaction among participants compared to desktop virtual reality. Hence, immersive VR proves to be more effective in engendering a flow experience than desktop VR.

6. Discussion

This study aimed to investigate the impact of immersive virtual reality environments on the flow experience of law school students, specifically in the realm of information technology law (IT law - Cyberlaw). To achieve this, two distinct virtual court environments were developed, real-life scenarios for clinical legal education were meticulously designed, and these scenarios were subsequently translated into immersive virtual reality settings, where case practices were conducted with law students. Employing an experimental design, the research facilitated comparisons between immersive virtual reality and desktop virtual reality environments concerning the flow experience. The findings indicate that participants in the immersive virtual reality group exhibited significantly higher levels of focused attention, telepresence, time distortion, and interaction compared to their counterparts in the desktop virtual reality group, thereby experiencing a more pronounced state of flow. Notably, the sole variable that did not exhibit a significant difference between the two environments was enjoyment.

In their research, Cheng et al. (2014) employed an interactive virtual reality environment and applied structural equation modeling to investigate the factors influencing the flow experience. Their findings indicated that within the virtual environment, the degree of interaction and the sense of presence (comprising concentration and time distortion) significantly impact the flow experience. Moreover, they identified that user skills act as a mediating variable, influencing the relationship between interaction and the flow experience. This aligns with the findings of Choi and Baek (2011), who emphasized the pivotal role of interaction within the desktop virtual reality environment in shaping the felt flow experience. To achieve a state of flow in a learning environment, learners must be able to concentrate on activities and actively interact. In essence, the potency of the flow experience within learning environments is intricately linked to the quality of interaction provided within the environment.

The results of the study indicate that law students who engaged with the immersive virtual reality group exhibited enhanced focus during case practice and achieved a state of effective immersion within the virtual court environment, effectively disconnecting from the constraints of time and space. Furthermore, the findings suggest that the immersive virtual reality environment fosters more efficient interaction among users. These outcomes are in line with the research conducted by Pallavicini et al. (2019), which compared the flow experience between immersive and desktop virtual reality environments and found that users in immersive settings reported a significantly higher flow experience. Similarly, other studies have consistently shown that users immersed in virtual reality environments experience a stronger sense of presence compared to those in desktop virtual reality settings (Makransky et al., 2019; Makransky & Lilleholt, 2018). These findings

Table 5
Normal distribution (skewness and kurtosis) results.

Variable	Skewness	Kurtosis
Focused Attention	-,410	-,447
Telepresence	-,821	-,061
Time Distortion	-,669	,432
Enjoyment	−1254	1519
Interaction	-,813	,583

Table 6
Correlation matrix.

n = 83	Focused Attention	Telepresence	Time Distortion	Enjoyment	Interaction
Focused Attention	1	,811[a]	,526[a]	,592[a]	,472[a]
Telepresence	,811[a]	1	,571[a]	,768[a]	,539[a]
Time Distortion	,526[a]	,571[a]	1	,579[a]	,614[a]
Enjoyment	,592[a]	,768[a]	,579	1	,674[a]
Interaction	,472[a]	,539[a]	,614[a]	,674[a]	1

[a] Correlation is significant at the 0.01 level (2-tailed).

Table 7
MANOVA (multivariate analysis of variance) results.

Pillai's Trace	F	Hypothesis df	Error df	p	η^2	Power
,175	3269	5	77	,010	,175	,872

Table 8
ANOVA (analysis of variance) results.

Variable	SS	df	MS	F	p	η^2	Power
Focused Attention	11,279	1	11,279	12,756	,001	,136	,942
Telepresence	9,623	1	9623	10,082	,002	,111	,880
Time Distortion	6412	1	6412	8262	,005	,093	,811
Enjoyment	3701	1	3701	5220	,025	,061	,617
Interaction	5119	1	5119	9201	,003	,102	,850

Table 9
Pairwise comparison results.

Variable	(I) Virtual Reality	(J) Virtual Reality	$\Delta \overline{X}_{(I-J)}$	Se	p
Focused Attention	Desktop	Immersive	-,737	,206	,001
Telepresence	Desktop	Immersive	-,681	,214	,002
Time Distortion	Desktop	Immersive	-,556	,193	,005
Interaction	Desktop	Immersive	-,497	,164	,003

substantially corroborate the results obtained in this study.

However, the lack of a significant difference in the enjoyment dimension of the flow experience comes as an unexpected outcome. Bravo et al. (2019) conducted research where they explored students' perceptions of interactive virtual cases, and many students expressed their appreciation for the enjoyable and positively engaging experiences provided by the virtual environment. Likewise, in several educational studies (Seybert et al., 2008; Smith et al., 2014), students have reported finding virtual environment practices to be enjoyable and satisfying. Furthermore, Reer et al. (2022) and Makransky and Lilleholt (2018) both demonstrated, in their respective studies, that immersive virtual reality engendered a stronger sense of enjoyment compared to non-immersive virtual reality. Consequently, the results related to the enjoyment factor appear to be at odds with the broader literature on the subject. It's possible that the content and design of the clinical law practices in both environments were equally engaging and well-crafted, leading to similar levels of enjoyment for the students. The primary focus of clinical law practices is to develop legal skills, knowledge, and competencies. If both environments effectively facilitated the achievement of these learning objectives, students might have derived a similar level of satisfaction regardless of the technology used. On the other hand, students may have varying preferences and comfort levels with different types of technology. Some students might have found the immersive VR experience enjoyable, while others preferred the desktop VR environment. These individual differences can balance out in the overall results. Additionally, students today are increasingly familiar with technology, including VR. They may have adapted quickly to both immersive and desktop VR environments, making the technology itself less of a novelty and more of a tool to achieve their educational goals. In summary, the non-significant difference in enjoyment between the immersive and desktop VR environments for clinical law practices may be the result of a combination of factors related to the design of the experience and individual preferences. Further research and a deeper analysis of participant feedback may provide additional insights into this phenomenon.

7. Conclusion and implications

In this study, a unique approach was taken to design and transfer clinical legal cases in various IT law fields into two distinct virtual environments: immersive and desktop. This allowed law students to engage with these cases as if they were real-life scenarios. Subsequently, the study compared the experiences of two groups, one using immersive VR and the other using desktop VR, with a focus on the flow experience dimensions, including focused attention, telepresence, time distortion, enjoyment, and interaction. To the best of the authors' knowledge, this research marks the first instance of such an approach in the field of law, making it a pioneering contribution to the domain.

There are various theoretical and practical inferences that can be made based on the empirical evidence obtained in the study. Immersion enhances both learning and engagement (Bodzin et al., 2020; Dede, 2009). Our findings indicate that a more immersive virtual environment enhances the flow experience. Therefore, it can be concluded that our findings will enhance learning outcomes in legal education by incorporating the immersion element, which is a key aspect of the flow experience. Study findings support the notion that immersive VR environments can enhance flow and cognitive engagement. Immersion in a virtual court setting likely led to greater concentration and attention to the case practice, which can positively affect learning outcomes. The sense of being isolated from time and space in immersive VR is a critical concept in VR research. This study provides empirical support for the idea that higher levels of presence can lead to better task performance and engagement. The suggestion that the immersive VR environment offers more effective interaction underscores the importance of designing VR learning experiences that leverage the unique affordances of this technology. Researchers and educators can explore how to optimize interaction design in immersive VR to enhance learning.

Immersive VR can be a valuable tool for enhancing the learning experience in legal education. Law schools and institutions can consider integrating immersive VR into their curriculum to provide students with more engaging and effective learning opportunities for case practice and courtroom simulations. Immersive VR can potentially offer flexibility in legal education by allowing students to access virtual courtrooms and practice sessions remotely. This can be especially beneficial for distance learning or students with scheduling constraints. The immersive VR environment can facilitate the development of practical skills that are crucial for legal professionals. Law students can benefit from realistic, immersive courtroom experiences that prepare them for real-world legal practice. Institutions and VR developers may be encouraged to invest in the further development of legal VR simulations and environments. For instance, Doğan and Şahin (2024) propose a set of evidence-based instructional design principles for utilizing immersive VR in clinical education settings. These principles can be adapted for use in clinical law education as well. This includes creating more sophisticated and content-rich virtual courtrooms that align with specific legal practice areas. Beyond law schools, legal practitioners and professionals can use immersive VR for ongoing training and professional development. It can provide a safe and immersive space for practicing legal scenarios and refining courtroom skills. Collaborations between legal experts, educators, and VR specialists may become more common to design and implement effective legal VR experiences. This interdisciplinary approach can lead to innovative solutions for legal education.

Several promising avenues for future research can be identified in this study's context. In the field of law, there is a need for investigations that delve deeper into how immersive VR impacts not only sensory aspects but also cognitive processes. Furthermore, it is crucial to conduct in-depth analyses of the enjoyment variable in both immersive VR and desktop VR settings within law clinics to understand its influence on the flow experience comprehensively. To enhance the generalizability of findings, future research could explore larger samples and diverse contexts. While this study focused on the application of immersive VR and desktop VR within clinical law courses, there remains a pertinent question of how similar virtual environments perform in theory-based instructional methods, a topic deserving significant research attention. Finally, expanding the utilization of these developed virtual environments as instructional tools, not only in law clinic courses but also during post-undergraduate internships, could significantly contribute to their widespread adoption and impact in legal education.

CRediT authorship contribution statement

Ezgi Doğan: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ferhan Şahin:** Writing – original draft, Methodology, Conceptualization. **Yusuf Levent Şahin:** Supervision, Software, Resources, Investigation. **Kadriye Kobak:** Writing – review & editing, Investigation. **Muhammet Recep Okur:** Writing – review & editing, Funding acquisition.

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